

Circular Motion

Single Correct Type Questions

1. A body is moving with constant speed, in a circle of radius 10m. The body completes one revolution in 4s. At the end of 3rd second, the displacement of body (in m) from its starting point is: [31 Jan, 2023 (Shift-II)]

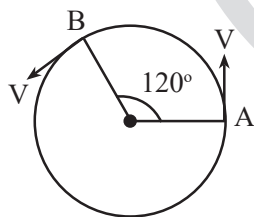
(a) 30 (b) 15π
(c) 5π (d) $10\sqrt{2}$

2. A particle is moving with constant speed in a circular path. When the particle turns by an angle 90° , the ratio of instantaneous velocity to its average velocity is $\pi : x\sqrt{2}$. The value of x will be [6 April, 2023 (Shift-I)]

(a) 2 (b) 5
(c) 1 (d) 7

3. As shown in the figure, a particle is moving with constant speed π m/s. Considering its motion from A to B, the magnitude of the average velocity is:

[6 April, 2023 (Shift-II)]



(a) π m/s (b) $\sqrt{3}$ m/s
(c) $2\sqrt{3}$ m/s (d) $1.5\sqrt{3}$ m/s

4. An object moves at a constant speed along a circular path in a horizontal plane with centre at the origin. When the object is at $x = +2m$, its velocity is $-4\hat{j}$ m/s. The object's velocity (v) and acceleration (a) at $x = -2m$ will be

[29 Jan, 2023 (Shift-II)]

(a) $v = 4\hat{i}$ m/s, $a = 8\hat{j}$ m/s²
(b) $v = 4\hat{j}$ m/s, $a = 8\hat{i}$ m/s²
(c) $v = -4\hat{j}$ m/s, $a = 8\hat{i}$ m/s²
(d) $v = -4\hat{i}$ m/s, $a = -8\hat{j}$ m/s²

5. A ball is spun with angular acceleration $\alpha = 6t^2 - 2t$ where t is in second and α is in rads^{-1} . At $t = 0$, the ball has angular velocity of 10 rads^{-1} and angular position of 4 rad. The most appropriate expression for the angular position of the ball is: [28 June, 2022 (Shift-II)]

(a) $\frac{3}{2}t^4 - t^2 + 10t$ (b) $\frac{t^4}{2} - \frac{t^3}{3} + 10t + 4$
(c) $\frac{2t^4}{3} - \frac{t^3}{6} + 10t + 12$ (d) $2t^4 - \frac{t^3}{2} + 5t + 4$

6. Motion of a particle in $x - y$ plane is described by a set of following equations $x = 4 \sin\left(\frac{\pi}{2} - \omega t\right)$ m and $y = 4 \sin(\omega t)$ m.

The path of the particle will be:

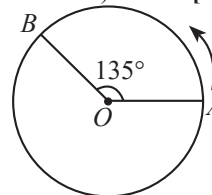
[28 June, 2022 (Shift-I)]

(a) Circular (b) Helical
(c) Parabolic (d) Elliptical

7. A person moved from A to B on a circular path as shown in figure. If the distance travelled by him is 60 m, then the magnitude of displacement would be:

(Given $\cos 135^\circ = -0.7$)

[25 July, 2022 (Shift-I)]



(a) 42 m (b) 47 m (c) 19 m (d) 40 m

8. A fly wheel is accelerated uniformly from rest and rotates through 5 rad in the first second. The angle rotated by the fly wheel in the next second, will be:

[24 June, 2022 (Shift-II)]

(a) 7.5 rad (b) 15 rad (c) 20 rad (d) 30 rad

9. A block of 200 g mass moves with a uniform speed in a horizontal circular groove, with vertical side walls of radius 20cm. If the block takes 40 s to complete one round, the normal force by the side walls of the groove is:

[16 March, 2021 (Shift-I)]

- (a) $6.28 \times 10^{-3} \text{ N}$ (b) 0.0314 N
(c) $9.859 \times 10^{-2} \text{ N}$ (d) $9.859 \times 10^{-4} \text{ N}$

10. A child of mass 5kg is going round a merry-go-round that makes 1 rotation in 3.14 s. The radius of the merry-go-round is 2 m. The centrifugal force on the child will be

[6 April 2023 (Shift-II)]

- (a) 80 N (b) 50 N
(c) 100 N (d) 40 N

11. A coin placed on a rotating table just slips when it is placed at a distance of 1 cm from the center. If the angular velocity of the table is halved, it will just slip when placed at a distance of from the centre: [11 April, 2023 (Shift-I)]

- (a) 2 cm (b) 1 cm (c) 8 cm (d) 4 cm

12. A car is moving on a horizontal curved road with radius 50m. The approximate maximum speed of car will be, if friction coefficient between tyres and road is 0.34. [Take $g = 10 \text{ ms}^{-2}$] [29 Jan, 2023 (Shift-I)]

- (a) 3.4 ms^{-1} (b) 22.4 ms^{-1}
(c) 13 ms^{-1} (d) 17 ms^{-1}

13. A body of mass 200 g is tied to a spring of spring constant 12.5 N/m , while the other end of spring is fixed at point O . If the body moves about O in a circular path on a smooth horizontal surface with constant angular speed 5 rad/s , then the ratio of extension in the spring to its natural length will be: [24 Jan, 2023 (Shift-II)]

- (a) 1 : 2 (b) 1 : 1 (c) 2 : 3 (d) 2 : 5

14. A vehicle of mass 200 kg is moving along a levelled curved road of radius 70 m with angular velocity of 0.2 rad/s . The centripetal force acting on the vehicle is :

[13 April, 2023 (Shift-II)]

- (a) 560 N (b) 2800 N
(c) 14 N (d) 2240 N

15. A car is moving with a constant speed of 20 m/s in a circular horizontal track of radius 40 m. A bob is suspended from the roof of the car by a massless string. The angle made by the string with the vertical will be : (Take $g = 10 \text{ m/s}^2$) [25 Jan, 2023 (Shift-I)]

- (a) $\frac{\pi}{6}$ (b) $\frac{\pi}{2}$
(c) $\frac{\pi}{4}$ (d) $\frac{\pi}{3}$

16. A disc with a flat small bottom beaker placed on it at a distance R from its center is revolving about an axis passing through the center and perpendicular to its plane with an angular velocity ω . The coefficient of static friction between the bottom of the beaker and the surface of the disc is μ . The beaker will revolve with the disc if:

[25 June, 2022 (Shift-II)]

- (a) $R \leq \frac{\mu g}{2\omega^2}$
(b) $R \leq \frac{\mu g}{\omega^2}$
(c) $R \geq \frac{\mu g}{2\omega^2}$
(d) $R \geq \frac{\mu g}{\omega^2}$

17. One end of a massless spring of spring constant k and natural length l_0 is fixed while the other end is connected to a small object of mass m lying on a frictionless table. The spring remains horizontal on the table. If the object is made to rotate at an angular velocity ω about an axis passing through fixed end, then the elongation of the spring will be: [27 June, 2022 (Shift-II)]

- (a) $\frac{k - m\omega^2 l_0}{m\omega^2}$ (b) $\frac{m\omega^2 l_0}{k + m\omega^2}$
(c) $\frac{m\omega^2 l_0}{k - m\omega^2}$ (d) $\frac{k + m\omega^2 l_0}{m\omega^2}$

18. A boy ties a stone of mass 100 g to the end of 2 m long string and whirls it around in a horizontal plane. The string can withstand the maximum tension of 80 N. If the maximum speed with which the stone can revolve is $\frac{K}{\pi} \text{ rev./min}$. The value of K is:

(Assume the string is massless and unstretchable)

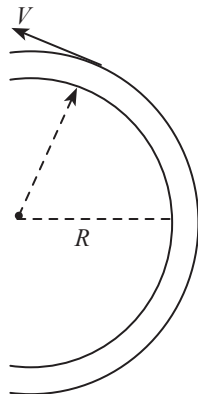
[24 June, 2022 (Shift-I)]

- (a) 400 (b) 300 (c) 600 (d) 800

19. A particle is moving with uniform speed along the circumference of a circle of radius R under the action of a central fictitious force F which is inversely proportional to R^3 . Its time period of revolution will be given by: [26 Feb, 2021 (Shift-I)]

- (a) $T \propto R^{\frac{3}{2}}$ (b) $T \propto R^{\frac{5}{2}}$
(c) $T \propto R^{\frac{4}{3}}$ (d) $T \propto R^2$

20. A modern grand - prix racing car of mass m is traveling on a flat track in a circular arc of radius R with a speed v . If the coefficient of static friction between the tyres and the track is μ_s , then the magnitude of negative lift F_L acting downwards on the car is :
(Assume forces on the four tyres are identical and g = acceleration due to gravity)[17 March, 2021 (Shift-I)]



- (a) $m\left(\frac{v^2}{\mu_s R} - g\right)$ (b) $-m\left(g + \frac{v^2}{\mu_s R}\right)$
(c) $m\left(g - \frac{v^2}{\mu_s R}\right)$ (d) $m\left(\frac{v^2}{\mu_s R} + g\right)$
21. A particle of mass m is suspended from a ceiling through a string of length L . The particle moves in a horizontal circle of radius r such that $r = \frac{L}{\sqrt{2}}$. The speed of particle will be:
[26 Aug, 2021 (Shift-II)]

- (a) $\sqrt{rg}$ (b) $\sqrt{\frac{rg}{2}}$
(c) $2\sqrt{rg}$ (d) $\sqrt{2rg}$

22. A particle of mass m is fixed to one end of light spring having force constant k and unstretched length ℓ . The other end is fixed. The system is given an angular speed ω about the fixed end of the spring such that it rotates in a circle in gravity free space. Then the stretch in the spring is:
[8 Jan, 2020 (Shift-I)]

- (a) $\frac{m\ell\omega^2}{k - m\omega^2}$ (b) $\frac{m\ell\omega^2}{k - \omega m}$
(c) $\frac{m\ell\omega^2}{k + m\omega^2}$ (d) $\frac{m\ell\omega^2}{k + m\omega}$

Integer Type Questions

23. A car is moving on a circular path of radius 600 m such that the magnitudes of the tangential and centripetal acceleration are equal. Time taken by the car to complete first quarter of revolution, if it is moving with an initial speed of 54 km/hr is $t(1 - e^{-\pi/2})$ s. The value of t is _____.
[29 Jan, 2023 (Shift-II)]
24. A body rotating with an angular speed of 600rpm is uniformly accelerated to 1800rpm in 10 sec. The number of rotations made in the process is _____.
[20 July, 2021 (Shift-II)]
25. A stone tied to 180 cm long string at its end is making 28 revolutions in horizontal circle in every minute. The magnitude of acceleration of stone is $\frac{1936}{x} \text{ ms}^{-2}$. The value of x _____.
(Take $\pi = \frac{22}{7}$) [30 Jan, 2023 (Shift-II)]
26. A curved in a level road has a radius 75 m. The maximum speed of a car turning this curved road can be 30 m/s without skidding. If radius of curved road is changed to 48m and the coefficient of friction between the tyres and the road remains same, then maximum allowed speed would be _____ m/s.
[25 June, 2022 (Shift-II)]

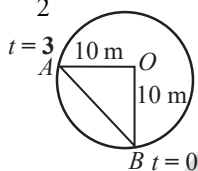
ANSWER KEY

- | | | | | | | | | | |
|---------|---------|----------|-----------|-----------|----------|---------|---------|---------|---------|
| 1. (d) | 2. (a) | 3. (d) | 4. (b) | 5. (b) | 6. (a) | 7. (b) | 8. (b) | 9. (d) | 10. (d) |
| 11. (d) | 12. (c) | 13. (c) | 14. (a) | 15. (c) | 16. (b) | 17. (c) | 18. (c) | 19. (d) | 20. (a) |
| 21. (a) | 22. (a) | 23. [40] | 24. [200] | 25. [125] | 26. [24] | | | | |

EXPLANATIONS

1. (d) Angle rotated by the moving body

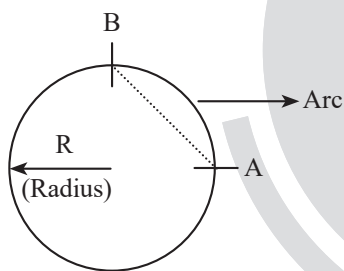
$$\theta = \frac{3}{4} \times 2\pi = \frac{3\pi}{2}$$



$$|\text{Displacement}| = AB = \sqrt{2}R$$

$$= 10\sqrt{2} \text{ m}$$

2. (a)



$$\text{Arc length} = 2\pi \left(\frac{R}{4} \right)$$

$$= \frac{\pi R}{2}$$

Let instantaneous velocity be v .

Time to cover arc length

$$t = \frac{\text{Arc length}}{v} = \frac{\pi R}{2v}$$

$$AB = R\sqrt{2}$$

Average velocity,

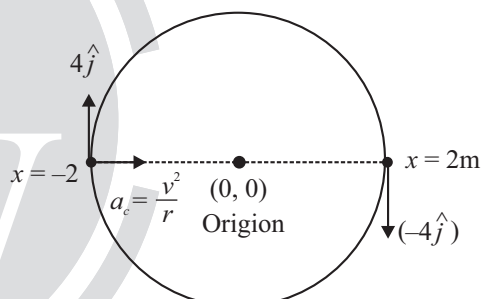
$$\langle v \rangle = \frac{AB}{t} = \frac{R\sqrt{2}(2v)}{\pi R} = \frac{2\sqrt{2}v}{\pi}$$

$$\Rightarrow \frac{v}{\langle v \rangle} = \frac{\pi}{2\sqrt{2}}$$

$$3. (d) \text{ Average velocity} = \frac{V \sin\left(\frac{\theta}{2}\right)}{\left(\frac{\theta}{2}\right)} = \frac{2V \sin\left(\frac{\theta}{2}\right)}{\theta}$$

$$= \frac{2V \sin 60^\circ}{\frac{2\pi}{3}} = 3 \sin 60^\circ = 1.5\sqrt{3} \text{ m/s}$$

4. (b)



Centripetal acceleration is given by

$$a_c = \frac{v^2}{r} = \frac{4^2}{2} = 8 \text{ m/s}^2$$

Using the above diagram

$$\vec{v} = 4\hat{j}$$

$$\text{and, } \vec{a}_c = 8\hat{i}$$

$$5. (b) \alpha = \frac{d\omega}{dt}$$

$$\Rightarrow \int_{10}^{\omega} d\omega = \int_0^t \alpha dt$$

$$\Rightarrow [\omega]_{10}^{\omega} = \int_0^t \alpha dt$$

$$\Rightarrow \omega - 10 = \int_0^t (6t^2 - 2t) dt$$

$$\omega = 10 + 2t^3 - t^2$$

EXPLANATIONS

$$\omega = 2t^3 - t^2 + 10$$

$$\frac{d\theta}{dt} = 2t^3 - t^2 + 10$$

$$\Rightarrow \int_4^{\theta} d\theta = \int_0^t (2t^3 - t^2 + 10) dt$$

$$\Rightarrow \theta - 4 = \frac{t^4}{2} - \frac{t^3}{3} + 10t$$

$$\therefore \theta = \frac{t^4}{2} - \frac{t^3}{3} + 10t + 4$$

$$6. (a) x = 4 \sin\left(\frac{\pi}{2} - \omega t\right) = 4 \cos(\omega t) \quad \dots(i)$$

$$y = 4 \sin(\omega t) \quad \dots(ii)$$

From (i) and (ii)

$$\cos^2 \omega t + \sin^2 \omega t = \left(\frac{x}{4}\right)^2 + \left(\frac{y}{4}\right)^2 = 1$$

$$= x^2 + y^2 = (4)^2$$

$$7. (b) \text{ Distance moved, } AB = 60 \text{ m}$$

$$\therefore (\theta) = \frac{\text{Arc}(AB)}{\text{radius}(R)} = \frac{3\pi}{4} = \frac{60}{R}$$

$$R = \frac{60 \times 4}{3\pi} = \frac{80}{\pi} \text{ m}$$

$$\text{Displacement} = \sqrt{R^2 + R^2 - 2R^2 \cos 135}$$

$$\{\because \cos 135 = \cos(90 + 45) = \sin 45^\circ = -0.7\}$$

$$= \sqrt{2R^2 - 2R^2(-0.7)}$$

$$= \sqrt{3.4R^2}$$

$$= \sqrt{3.4 \left(\frac{80}{4}\right)^2}$$

$$\approx 47 \text{ m}$$

$$8. (b) \theta_1 = \frac{1}{2} \alpha (1)^2 = 5 \text{ rad}$$

$$\Rightarrow \alpha = 10 \text{ rad/sec}^2$$

$$\text{So } \theta_2 = \frac{1}{2} \times 10 \times 2 \times 2 = 20 \text{ rad}$$

$$\therefore \theta_2 - \theta_1 = 20 - 5 = 15 \text{ rad}$$

$$9. (d) \text{ The normal force by the side walls of the groove,}$$

$$F = m\omega^2 R = m \left(\frac{2\pi}{T}\right)^2 R = \frac{0.2 \times 4 \times \pi^2 \times 0.2}{40 \times 40}$$

$$= 9.859 \times 10^{-4} \text{ N}$$

$$10. (d) \omega = \frac{2\pi}{T} = \frac{2\pi}{3.14} = 2 \text{ rad/s}$$

$$F_{\text{centrifugal}} = M\omega^2 R = 5 \times (2)^2 \times 2 = 40 \text{ N}$$

$$11. (d) \text{ Frictional force would balance the centrifugal force i.e.,}$$

$$f_{s \text{ max}} = m\omega^2 R$$

$$\mu mg = m\omega^2 R$$

$$\Rightarrow \text{So when } \omega \text{ becomes } \frac{\omega}{2}, R \text{ will become } 4R.$$

$$\Rightarrow \text{Therefore distance from the center will be 4 cm.}$$

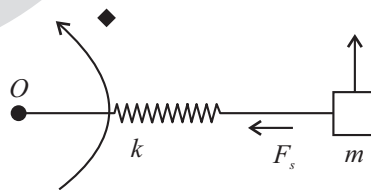
$$12. (c) \text{ For maximum speed in safe turning,}$$

$$\frac{mV_{\text{max}}^2}{r} = \mu mg$$

$$\Rightarrow V_{\text{max}} = \sqrt{\mu rg}$$

$$= \sqrt{0.34 \times 50 \times 10} \approx 13 \text{ m/s}$$

$$13. (c)$$



Let, natural length of the spring = L_0

Extension in the spring = x

$$\Rightarrow kx = m(L_0 + x)\omega^2$$

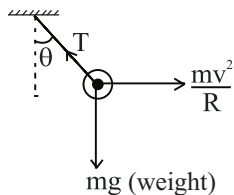
$$\Rightarrow 12.5x = \frac{1}{5}(L_0 + x)25$$

$$\Rightarrow 1.5x = L_0$$

$$\Rightarrow \frac{x}{L_0} = \frac{2}{3}$$

$$14. (a) \text{ Centripetal Force, } F_c = m\omega^2 r = 200 \times (0.2)^2 \times 70 = 560 \text{ N}$$

15. (c)



Equating the horizontal and vertical forces,

$$T \cos \theta = mg$$

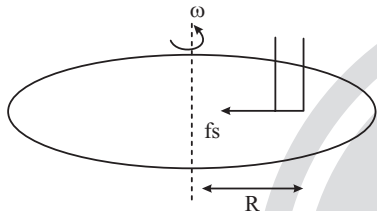
$$T \sin \theta = \frac{mv^2}{R}$$

$$\Rightarrow \tan \theta = \frac{v^2}{Rg} = \frac{20^2}{40 \times 10}$$

$$\Rightarrow \tan \theta = 1$$

$$\therefore \theta = \frac{\pi}{4}$$

16. (b) For beaker to move with discs



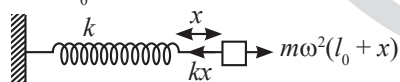
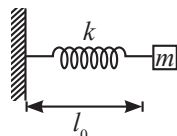
$$f_s = m\omega^2 R$$

We know that $f_s \leq f_{s,\max}$

$$m\omega^2 R \leq \mu mg$$

$$R \leq \frac{\mu g}{\omega^2}$$

17. (c)



$$\Rightarrow kx = m\omega^2(l_0 + x)$$

$$kx - m\omega^2 l_0 - m\omega^2 x = 0$$

$$x = \frac{m\omega^2 l_0}{k - m\omega^2}$$

18. (c) $T = m\omega^2 r$

$$\Rightarrow 80 = 0.1 \times \left(2\pi \times \frac{K}{\pi} \times \frac{1}{60} \right)^2 \times 2$$

$$\Rightarrow \frac{800}{2} = \frac{K^2}{900}$$

$$\Rightarrow K = 30 \times 20 = 600$$

19. (d) $F = \frac{k}{R^3} = \frac{mv^2}{R}$ (here, k = constant)

$$\Rightarrow v = \sqrt{\frac{k}{mR^2}}$$

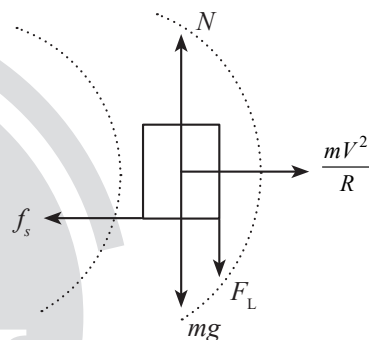
$$T = \frac{2\pi R}{V} = \frac{2\pi R}{\sqrt{\frac{k}{m} \times \left(\frac{1}{R}\right)}} \Rightarrow T \propto R^2$$

20. (a) $N = mg + F_L$

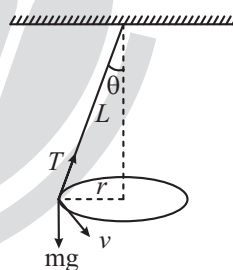
$$f_s \leq \mu_s \cdot N$$

$$\therefore \frac{mv^2}{R} \leq \mu_s N$$

$$\frac{mv^2}{R} = \mu_s (mg - F_L) \Rightarrow -F_L = m \left(\frac{v^2}{\mu_s R} - g \right)$$



21. (a)



Given, radius of horizontal circle, $r = \frac{L}{\sqrt{2}}$

Using free body diagram,

$$\sin \theta = \frac{r}{L} = \frac{1}{\sqrt{2}} \Rightarrow \theta = 45^\circ$$

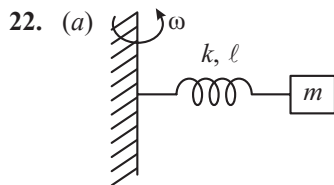
$$T \sin \theta = \frac{mv^2}{r} \quad \dots (i)$$

$$T \cos \theta = mg \quad \dots (ii)$$

On dividing (i) with (ii), we get,

$$\tan \theta = \frac{v^2}{rg}$$

$$\Rightarrow v = \sqrt{rg} \quad [\because \theta = 45^\circ]$$



$$m\omega^2(\ell + x) = kx; \left(\frac{\ell}{x} + 1\right) = \frac{k}{m\omega^2}$$

$$x = \frac{\ell m\omega^2}{k - m\omega^2}$$

23. [40] $\because a_c = \frac{v^2}{R}$

$$v \frac{dv}{dx} = \frac{v^2}{R}$$

$$\Rightarrow \int_{15}^v \frac{dv}{v} = \frac{1}{R} \int_0^x dx$$

$$\Rightarrow v = 15e^{x/R}$$

$$\because \frac{dx}{dt} = 15e^{x/R} \quad \left[\because v = \frac{dx}{dt} \right]$$

arranging the terms and integrating both sides

$$\int_0^{\frac{\pi R}{2}} e^{-x/R} dx = 15 \int_0^{t_0} dt$$

$$\Rightarrow t_0 = 40(1 - e^{-\pi/2})$$

24. [200] $\omega^2 = \omega_0^2 + 2\alpha\theta \Rightarrow \left(\frac{2\pi}{60} \times 1800\right)^2$

$$= \left(\frac{2\pi}{60} \times 600\right)^2 + 2\left(\frac{2\pi}{60} \times \frac{1200}{10}\right)0$$

On solving, $\theta = \frac{80\pi \times 40\pi}{2 \times 4\pi} = 400\pi \text{ rad}$

Number of rotations = $\frac{\theta}{2\pi} = \frac{400\pi}{2\pi} = 200$

25. [125] As the stone is moving in a circular path, so the

acceleration is $a = \omega^2 R = \left(\frac{28 \times 2\pi}{60}\right)^2 \times 1.8$

$$= \frac{1936 \times 1.8}{225}$$

$$x = 125$$

26. [24] $f_{s \max} = \frac{mv^2}{R}$

$$\mu mg = \frac{mv^2}{R}$$

$$v = \sqrt{\mu Rg}$$

$$\frac{v_2}{v_1} = \sqrt{\frac{R_2}{R_1}}$$

$$\frac{v_2}{30} = \sqrt{\frac{48}{75}}$$

$$v_2 = 24 \text{ m/s}$$